AWS VPC PEERING

Student Project Report

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Project	AWS VPC Peering — Connect Two VPCs
AWS Region	Asia Pacific (Sydney) — ap-southeast-2
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Project objective

Design two non-overlapping Amazon Virtual Private Clouds (VPCs), establish a VPC peering connection, configure route tables and security-group rules, and provide private communication paths between application workloads.

Report scope

This report documents the architecture and AWS console configuration observed, alongside the required network design and validation method from the EWB Cloud Labs workbook. It does not claim that connectivity tests succeeded unless actual test results have been captured.

Based on the EWB Cloud Labs Student Project Workbook — AWS VPC Peering, Project 01 (October 2026).

1. Architecture and Network Design

The target architecture uses VPC A for a public bastion and a private application workload. VPC B contains a private application workload and has no internet gateway. The VPCs communicate through a direct peering connection using private IPv4 addresses.

Resource	CIDR / target address	Observed information
ewb-vpc-a	10.10.0.0/16	VPC ID: vpc-08bb182cece02cde9
ewb-a-public	10.10.1.0/24	Subnet ID: subnet-0304424530ff4d2b5
ewb-a-private	10.10.2.0/24	Workbook-required subnet; not confirmed in shared evidence
ewb-vpc-b	10.20.0.0/16	VPC ID: vpc-0ac24b6980dca6643
ewb-b-private	10.20.1.0/24	Subnet ID: subnet-0d151553e9e776a73
ewb-bastion	Target 10.10.1.10; public IPv4	Observed private IP 10.10.1.99; public IP 3.27.222.24
ewb-app-a	Target 10.10.2.10; no public IP	Instance not confirmed in shared evidence
ewb-app-b	Target 10.20.1.10; no public IP	Instance not confirmed in shared evidence

Address consistency note

The workbook's example bastion address is 10.10.1.10, while the observed bastion private address is 10.10.1.99. Any security-group /32 source or SSH configuration must use the actual instance address, or all dependent settings must be updated consistently.

2. Resource Inventory

Resource	Observed ID / configuration
VPC A	ewb-vpc-a · vpc-08bb182cece02cde9 · 10.10.0.0/16
VPC B	ewb-vpc-b · vpc-0ac24b6980dca6643 · 10.20.0.0/16
Peering connection	ewb-vpc-peering · pcx-0245d1610d3b17c7b · Active
Public route table	ewb-rt-a-public · rtb-0f67b9c5b3ddf7515
Private route table B	ewb-rt-b-private · rtb-0f3775cfad1d31a53
Private route table A	Workbook target: ewb-rt-a-private; not confirmed in shared evidence
Bastion EC2	ewb-bastion · i-0dd78b8d62372de78 · Amazon Linux 2023 · t3.micro · Running
Bastion private/public IP	10.10.1.99 / 3.27.222.24
Bastion security group	ewb-sg-bastion · sg-0e720ca3d619eda2b
App-A security group	ewb-sg-app-a · sg-0c5ad150534472c82
App-B security group	ewb-sg-app-b · sg-0c47bbecc7fd6c406
Key pair	ewb-vpc-key (created; local .pem availability not verified)

3. Routing Configuration

The following routes were confirmed in the AWS console. The VPC A public route table and VPC B private route table both include routes through the active peering connection.

Route table	Destination	Target	Status
ewb-rt-a-public	10.10.0.0/16	local	Active
ewb-rt-a-public	0.0.0.0/0	igw-093f672ba5f7aa35b	Active
ewb-rt-a-public	10.20.0.0/16	pcx-0245d1610d3b17c7b	Active
ewb-rt-b-private	10.20.0.0/16	local	Active
ewb-rt-b-private	10.10.0.0/16	pcx-0245d1610d3b17c7b	Active
ewb-rt-a-private	10.10.0.0/16 local; 10.20.0.0/16 via pcx-ID	Workbook-required	Not confirmed

The workbook specifies three route tables. The private route table in VPC A must be associated with ewb-a-private and include the local route plus 10.20.0.0/16 targeting the peering connection. The public route table also needs the peering route because the bastion initiates connections to App-B.

4. Security Groups and Access Design

Security group	Required / observed inbound configuration	Review note
ewb-sg-bastion	Observed SSH TCP/22 source: 49.37.8.92/32	Confirm source is current public IPv4 /32
ewb-sg-app-a	Observed SSH from 10.10.1.10/32; ICMP from 10.20.1.10/32	Check actual host IPs; bastion observed as .99
ewb-sg-app-b	Workbook requires SSH from bastion private IP and ICMP Echo Request from App-A private IP	Actual rules not confirmed

Security principles

- Keep App-A and App-B private, with no public IPv4 addresses.
- Permit SSH only from the bastion's private address to application instances.
- Permit ICMP Echo Request only from the intended peer application host.
- Do not expose SSH or ICMP to 0.0.0.0/0.
- The workbook assumes default allow-all network ACLs; custom ACLs require return traffic rules.

5. Connectivity Validation

The workbook requires the following tests. Expected outcomes are shown as acceptance criteria, not as observed results.

Test	Command / action	Expected outcome
Baseline	From App-A before peering routes: ping -c 4 10.20.1.10	No replies
Forward path	From App-A after routes: ping -c 4 10.20.1.10	4 replies; 0% packet loss
Reverse path	From App-B: ping -c 4 10.10.2.10	4 replies; 0% packet loss
Private SSH	From laptop: ssh ewb-app-b via bastion ProxyJump	Shell on App-B
Isolation	Check EC2 details for App-A and App-B	No public IPv4 addresses
Controlled failure	Remove App-B ICMP Echo Request from App-A; test; restore rule and test again	Ping fails then resumes

A peering status of Active and active routes confirm the AWS configuration is present; they do not prove end-to-end traffic. Actual terminal output must be captured to claim successful connectivity. Browser-based EC2 Instance Connect to the bastion previously failed, so SSH access must be resolved before running these tests.

6. Administration and Troubleshooting

The workbook uses a local SSH client and ProxyJump. The private key remains on the student's computer and must not be copied to an EC2 instance or included in the report.

Example SSH configuration (replace the key path and IPs if your actual settings differ):

```
Host ewb-bastion
    HostName 3.27.222.24
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes

Host ewb-app-a
    HostName 10.10.2.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion

Host ewb-app-b
    HostName 10.20.1.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion
```

Troubleshooting order

1. Confirm instance state and destination IP.
2. Check the associated route table and both directions of the peering routes.
3. Verify security-group source IPs match actual host addresses.
4. For bastion SSH timeouts, check public IP, client IP /32 rule, Internet Gateway attachment and public route association.
5. A public-key authentication error usually indicates key or username configuration, not necessarily a routing failure.

7. Key Concepts

- VPC peering connects non-overlapping VPC CIDR ranges using private routes.
- Both sides need routes so request and reply traffic can flow.
- The bastion subnet needs a peering route to reach App-B through ProxyJump.
- Security groups are stateful; replies to permitted traffic are allowed automatically.
- VPC peering is non-transitive and does not give VPC B internet access through VPC A's Internet Gateway.
- The more-specific 10.20.0.0/16 route is chosen over the 0.0.0.0/0 route for destinations in VPC B.

8. Conclusion

The observed AWS configuration establishes an Active peering connection between ewb-vpc-a (10.10.0.0/16) and ewb-vpc-b (10.20.0.0/16), with active peering routes visible in the VPC A public route table and VPC B private route table. Full validation requires confirming the VPC A private route table, both private application instances, security-group rules, and real terminal evidence for baseline, two-way ping, private SSH and controlled-failure/recovery tests.

9. Reference

EWB Cloud Labs, *Student Project Workbook — AWS VPC Peering*, Project 01, October 2026; pages 3–13.

Screenshots omitted as requested. Add your own console and terminal screenshots to the GitHub repository or append them to a final evidence PDF. Never include private keys, passwords or access tokens.